1 SciFact corpus

no queries, no relevance labels

- Chronic pancreatitis is...
- Microstructural...
- Induction of...
- BC1 RNA, the transcript...

2 An LLM writes queries

from sampled documents, 100 per task

- What non-genetic factors can trigger cancer development in the presence of oncogenic mutations?
- Bcl-2's anti-apoptotic function depends on binding to Bax.

28 embedding models
retrieve their top 10

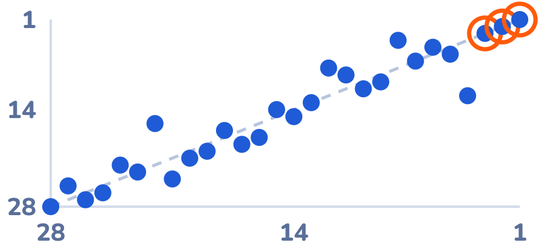
3 A judge grades the results

pointwise: each result 0 to 5
pairwise: which top 10 is better, A or B

A Model A grade 0–5		B Model B grade 0–5	
1	3.3	1	1.1
2	4.3	2	1.2
3	2.7	3	0.6
... 7 more		... 7 more	

vs

Agrees with MTEB ρ = 0.95



Same top 3 as MTEB

1. Qwen3-Embedding-8B
2. Qwen3-Embedding-4B
3. Linq-Embed-Mistral

4 Gym ranking

by nDCG@10 with the grades

	Gym MODEL	MTEB
1	Qwen3-Embedding-8B gym score 82.7	1
2	Qwen3-Embedding-4B gym score 81.6	2
3	Linq-Embed-Mistral gym score 81.2	3
4	harrier-oss-v1-0.6b gym score 80.2	8
5	gte-Qwen2-7B-instruct gym score 79.8	6
6	F2LLM-v2-4B gym score 79.6	5
7	jina-embeddings-v5-text... gym score 78.6	7

... 21 more